TEAM TASK TRACKER — SETUP AND RUNNING GUIDE

This document explains how to set up and run both Frontend and Backend on any computer from a fresh Git clone or ZIP file.

1. PREREQUISITES

Make sure the following are installed on the computer:

- **Python:** Version 3.10 or higher
- **Node.js:** Version 18.x or higher (includes npm)
- **MongoDB:** Local MongoDB service running on port 27017 or a MongoDB Atlas URI

2. CLONE OR EXTRACT PROJECT

Open terminal or Command Prompt and clone the repository:

```
git clone https://github.com/SAfsehEhsani/Team-Task-Tracker-Jira.git
cd Team-Task-Tracker-Jira
```

3. METHOD 1: 1-CLICK AUTOMATIC RUN (WINDOWS)

Double-click **start.bat** in the project root folder, or run in Command Prompt:

```
.\start.bat
```

What this script does automatically:

- Checks if MongoDB service is running.
- Checks if **backend_venv** exists. If not, it creates a new virtual environment and installs all packages from **backend\requirements.txt**.
- Checks if **frontend\node_modules** exists. If not, it runs **npm install**.
- Launches Django Backend on **http://127.0.0.1:8000**.
- Launches Next.js Frontend on **http://localhost:3000**.
- Automatically opens **http://localhost:3000** in your default web browser.

4. METHOD 2: STEP-BY-STEP MANUAL COMMANDS

If you want to run backend and frontend manually in separate terminal windows:

Step A: Start MongoDB Service

```
net start MongoDB
```

Step B: Terminal 1 — Backend (Django)

Run the following commands in the main project folder:

```
python -m venv backend_venv
backend_venv\Scripts\activate
pip install -r backend\requirements.txt
python backend\manage.py runserver 127.0.0.1:8000
```

Step C: Terminal 2 — Frontend (Next.js)

Open a new terminal window and run:

```
cd frontend
npm install
npm run dev
```

5. MAC / LINUX USERS

Mac and Linux users can run the shell script:

```
chmod +x start.sh
./start.sh
```

6. APPLICATION LINKS & DEMO ACCOUNTS

- **Frontend Web App:** http://localhost:3000
- **Backend API Health Check:** http://127.0.0.1:8000/api/health/

Pre-Configured Demo Accounts:

- **Administrator Account:**

- Email: **admin@teamtracker.local**

- Password: **Admin@12345**

- Access: Full control, team creation, member assignment, live MongoDB data inspector.

- **Team Member Account:**

- Email: **alice.chen@techcorp.io**

- Password: **Alice@12345**

- Access: View assigned tasks, update task status (To-Do, In Progress, In Review, Done).

- **New Account Registration:** Anyone can click **'Create an account'** on the login page to register a custom Administrator or Member account.

7. WHY BACKEND_VENV AND NODE_MODULES ARE NOT COMMITTED TO GITHUB

In Python and Node.js projects, virtual environments and **node_modules** are never uploaded to GitHub because they contain machine-specific hardcoded paths (e.g. **c:\users\...**) and compiled binary files for one specific operating system. If uploaded, they fail and crash immediately on any other computer. Instead, **backend/requirements.txt** and **frontend/package.json** are pushed to Git. When anyone clones the repository and runs **start.bat**, a clean, 100% compatible virtual environment and node_modules are automatically created on their computer without any errors.